

AI Reference Report

1. AI Tool Used

Tools: Claude Sonnet 4.5 / Claude Sonnet 4.6 (claude.ai + Claude Code CLI)

2. Scope of Usage — Claude Chat (claude.ai)

1. TiledMapCollider.ts (Lines: All)

Prompt: 幫我寫一個 TiledMapCollider，從 TMX 文字資產解析 CSV tile 資料，並用 run-length 合併相鄰 tile，替每段建立 PhysicsBoxCollider。

AI Response: Generated full TiledMapCollider.ts with TMX parsing, run-length encoding, and box.apply() calls.

Refinement: Added box.apply(). Fixed origin offset bug: ox/oy changed from -mapW/2,-mapH/2 to 0,0 — original values displaced all colliders 800px left and 120px down from visual tiles.

2. PlayerController.ts (Lines: All)

Prompt: 幫我寫 PlayerController，支援移動、跳躍、敵人踩踏、受傷無敵、死亡動畫，並用 Set 追蹤地面接觸。

AI Response: Generated full PlayerController.ts with Rigidbody physics, coyote time, contact listeners.

Refinement: Added _coyoteTimer (0.1s). Set col.friction=0. Added sprite atlas animation system. Added big Mario collision box resize with offset=(0,0) trick. Added shrink animation with SHRINK_SEQ. Fixed stomp: added isDead guard to prevent same-frame double contact. Added auto-jump (doJump()) on stomp. Fixed PTM to use cc.PhysicsManager.PTM_RATIO instead of hardcoded 32.

3. CameraFollow.ts (Lines: onLoad())

Prompt: 幫我寫 CameraFollow，掛在 Main Camera 上，用 clampf 限制地圖邊界，lateUpdate + lerp 平滑跟隨玩家。

AI Response: Generated CameraFollow.ts with target.x clamp and lerp smoothing.

Refinement: Added _snap() call in onLoad() to prevent first-frame position jump. HUD as child of Main Camera; GameWorld at (0,0) permanently.

4. GameUI.ts (Lines: All)

Prompt: 幫我寫 GameUI，顯示生命值、分數、倒數計時器(180 秒)，時間歸零呼叫 loseLife()。

AI Response: Generated GameUI.ts with livesLabel, scoreLabel, timerLabel, countdown logic.

Refinement: Removed 'TIME' prefix. Removed onDestroy null-setting to prevent killing new scene's listeners after reload. Added scheduleOnce 0-delay in onLoad.

5. GameManager.ts (Lines: All)

Prompt: 跨場景 Singleton GameManager，管理 lives、score、currentLevel，死亡時跳 GameOver 或重載場景。

AI Response: Generated GameManager.ts with addPersistRootNode, loseLife, addScore, loadLevel.

Refinement: Added isWin flag. Root-level node required for addPersistRootNode.

6. GameOverUI.ts (Lines: All)

Prompt: GameOverUI，顯示最終分數，勝利和失敗共用同一個 Scene。

AI Response: Generated GameOverUI.ts checking isWin to show 'GAME WIN!' or 'GAME OVER'.

Refinement: Added auto Firebase score submission. Button Click Events cleared.

7. Enemy.ts / QuestionBlock.ts (Lines: All)

Prompt: Enemy（巡邏/踩踏）和 QuestionBlock（從下方撞擊觸發，加 100 分）。

AI Response: Generated both scripts. QuestionBlock _activate() calls addScore(100).

Refinement: QuestionBlock hit detection changed to world Y comparison for reliability.

8. GoalFlag.ts (Lines: All (new file))

Prompt: 終點旗桿，碰到後切換到 GAME WIN，跟 GAME OVER 共用 Scene。加踩敵人音效和通關音效。

AI Response: Generated GoalFlag.ts with world X comparison, BGM stop, clearSfx play, 3-second delay before scene transition. Added stompSfx to PlayerController.

Refinement: Sets GameManager.isWin=true before loading GameOver scene. Delay uses clearSfx duration or 0.5s fallback.

9. PiranhaPlant.ts (Lines: All (new file))

Prompt: 新增花敵人，從水管鑽出來，收回時不能碰。部分花改為橫向往左移動。

AI Response: Generated PiranhaPlant.ts with state machine S_IDLE/S_RISING/S_OUT/S_SINKING. Sensor collider disabled when inside pipe. Added horizontal: boolean option.

Refinement: gravityScale=0 on Rigidbody. Position controlled via node.x/node.y directly.

10. Goomba.ts (Lines: All (new file))

Prompt: 新增 Goomba 敵人，用第一張圖靠 scaleX 翻轉走路，第二張為壓扁圖，踩死 +100 分。

AI Response: Generated Goomba.ts with single walkFrame + scaleX flipping, wall-reversal on horizontal contact.

Refinement: Fixed stomp detection: added isDead guard and other.enabled=false immediately on stomp to prevent same-frame double events. Inspector: PhysicsBoxCollider Offset Y adjusted upward.

11. FlyingGoomba.ts (Lines: All (new file))

Prompt: 新增飛天 Goomba，上下飛加往左飛，翅膀動畫，踩死 +300 分。

AI Response: Generated FlyingGoomba.ts with gravityScale=0, sine-wave vertical bob via linearVelocity, wing frame animation.

Refinement: Kinematic Rigidbody. On stomp, gravityScale restored to 1 for falling.

12. SuperMushroom.ts (Lines: All)

Prompt: 吃到蘑菇加 1000 分，蘑菇質量改成 0 讓它不能撞跑別的物件。

AI Response: Updated SuperMushroom.ts with addScore(1000). Set density=0.0001 after rb.enabled=true.

Refinement: sensor=true tried first but caused wall-clipping. Reverted to normal collider with density=0.0001. gravityScale=0 so mushroom floats while patrolling.

13. FirebaseManager.ts (Lines: All (new file))

Prompt: Firebase Manager，動態載入 SDK，支援 Auth、Firestore 排行榜和進度儲存。

AI Response: Generated FirebaseManager.ts as persistent singleton with ready Promise.

Refinement: Switched cc.loader.loadScript to native DOM createElement for reliable CDN loading. Added null guards. Filled in sshw2-f82d1 credentials.

14. LoginUI.ts (Lines: All (new file))

Prompt: 登入/註冊畫面，Email/Password，有 Guest 模式。

AI Response: Generated LoginUI.ts with _setMode, _onSubmit, _parseError, await ready.

Refinement: All text English (SIGN IN / SIGN UP / HAVEN'T SIGNED UP?). Added await ready at _onSubmit start. Fallback error shows actual error code.

15. LeaderboardUI.ts (Lines: All (new file))

Prompt: 排行榜畫面，動態建立節點顯示前 6 名，自己的那筆用橘色標示。

AI Response: Generated LeaderboardUI.ts dynamically creating row nodes.

Refinement: Changed from Layout+ResizeMode.CONTAINER (caused spacing issues due to deferred resize) to manual x-positioning with fixed row height=30px. All text black, current user orange.

16. MainMenuUI.ts (Lines: All)

Prompt: 更新 MainMenuUI，加排行榜按鈕、登出、登入狀態顯示。

AI Response: Updated with leaderboard/logout/login buttons and userLabel.

Refinement: Buttons toggle based on FirebaseManager.inst.isLoggedIn.

17. README.md (Lines: All (new file))

Prompt: 幫我寫 README，簡單明瞭，加上有哪些音效和動畫，用英文。

AI Response: Generated README.md with Features, Project Structure, Scene Flow, Sound Effects (7 SFX), Animations, Build & Deploy sections.

Refinement: Removed animation method column per user request. All content in English.

3. Scope of Usage — Claude Code (CLI)

1. TiledMapCollider.ts (Lines: onLoad(), _build())

Prompt: LEVEL1 按下播放之後腳色會往左噴射之後往下掉，並且穿模，我不知道為什麼。

AI Response: Diagnosed via Level1.fire JSON: (1) scheduleOnce 0.3s delay in onLoad() causes player to free-fall then colliders appear mid-fall triggering Box2D separation impulse. (2) ox/oy origin offset displaces colliders. Fixed both.

Refinement: Removed scheduleOnce — _build() now called directly in onLoad(). Also fixed CameraFollow.onLoad() to snap immediately.

2. PlayerController.ts (Lines: onLoad(), contacts, PTM)

Prompt: (同上診斷) 分析所有物理相關 bug。

AI Response: Diagnosed 5 bugs: missing fixedAngle, unreliable normal-based groundCount, PTM hardcoded as 32, normal sign inconsistency, scheduleOnce delay. Fixed top 3.

Refinement: Added rb.fixedAngle=true. Changed groundCount tracking to Set<cc.PhysicsContact> (add on begin, delete on end) for reliability. Changed PTM from hardcoded 32 to cc.PhysicsManager.PTM_RATIO.

3. PlayerController.ts (Lines: animation system)

Prompt: 我腳色的 walk animation 和 jump animation 在哪裡，怎麼好像沒看到。

AI Response: Analyzed mario_small.plist. Added programmatic atlas animation: ANIM_IDLE/WALK/JUMP constants, _updateAnim(dt) called from update().

Refinement: No .anim files in project. Used getSpriteFrame() from cc.SpriteAtlas. Frame names inferred from plist analysis.

4. Statement of Non-Usage

All code was either written by the student or generated/assisted by Claude Sonnet 4.5/4.6 as

documented above. No other AI tools were used. The student reviewed, tested, debugged, and refined all AI-generated code. Key decisions: Static physics bodies in fixed GameWorld, Main Camera outside Canvas for scrolling, HUD as Camera child, GameManager root-level node for addPersistRootNode, Firebase SDK loaded via DOM script injection.